

The Effect of Contamination on Contamination Control Regulation

US Coal Mining and Drinking Water Utilities

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Self-reported compliance

- Self-reported compliance: entity monitors and reports themselves to regulators.
- Incentives to shirk reporting:
 - ▶ Concealing non-compliance
 - ▶ Saving self-reporting costs
- Perversely, incentives to under-report increase with contamination for environmental contamination regulation (e.g. US drinking water).
- **Why care about self-reporting and contamination?**
 - ▶ Reporting is most valuable to regulators when contamination is rising
 - ▶ Different fix if concealing non-compliance vs self-reporting costs
 - ▶ 85% of US household drinking water is regulated this way
 - ▶ Self-reporting used across industries (HIPAA violation disclosure, SOX, energy firms)

Research question

1. What happens to firm self-reporting when contamination rises?
2. Are changes in self-reporting due to concealment or self-reporting costs?

Context: US drinking water utilities exposed to water contamination from upstream coal mining (1985–2005)

0. Does upstream coal mining raise contamination at drinking water utilities?

How I answer these questions

- Annual panel of drinking water utilities on average 22 km downstream of active coal mines between 1985 and 2005.
- Separation between contamination source and regulated firm helps identify the effect of contamination.
- Coal mining → inorganic chemicals (“IOCs”) into surface and ground water.
- Contaminant concentrations for several IOC’s 1998–2005 link coal production to contamination.
- Instrument upstream coal production.
- Utility self-reporting, regulator visits, and enforcement received.

Utility nitrate self-reporting and under-reporting incentives

Utility manager/staff cognitive burden (Oleckno, 1982).

Self-reporting requirement for nitrates (one of 83 contaminants):

- **Baseline:** once a year, send one sample from every intake to a lab
- **Elevated:** nitrate ≥ 5 mg/L, half the Maximum Contaminant Level (“MCL”), test quarterly

Miss a test, receive a monitoring and reporting (“MR”) violation and:

- Avoid an MCL (need self-report for MCL)
- MR punished less severely than MCL
- Regulator visit
- Or enforcement: Informal (letters, notices, compliance plans); Formal (consent decrees, fines, prosecution)

Is coal production a source of contamination at utilities?

$$Concen_{ct} = \beta \cdot \sum_{\tau=1985}^t \text{UpstreamCoalProd}_{c\tau} + \rho_c + \text{Watershed}_c \cdot \tau_t + \epsilon_{ct}$$

- $Concen_{ct}$: mean annual concentration at utility c , year t (SYR2, 1998–2005).
- β : effect of 1 million additional cumulative short tons mined upstream since 1985.
- ρ_c : utility fixed effects. $\text{Watershed}_c \cdot \tau_t$: Watershed-specific year effects, absorbing watershed-wide industrial trends.
- SEs clustered at the utility level.
- Continuous treatment, **no control**: identification comes from relative exposure.
- Assumption: utilities with large upstream production changes would have trended like those with small ones. (balance test on utility characteristics $\rightarrow$)

Mining raises contaminant levels

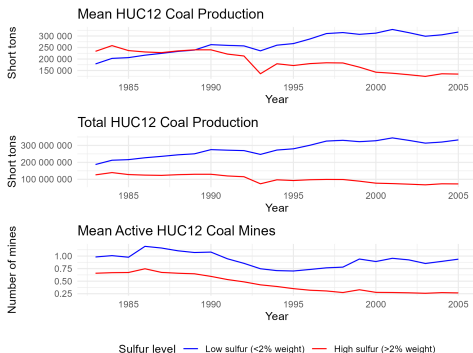
	Arsenic (mg/L) (1)	Nitrate (mg/L) (2)	Barium (mg/L) (3)	Selenium (mg/L) (4)
Cumul. upstream coal prod. (M ST)	0.00001 (0.00001)	0.00883*** (0.00257)	0.00136 (0.00112)	0.00004** (0.00002)
Sample mean (million short tons)	6.9	8.6	6.7	6.7
MCL (mg/L)	0.050	10.000	2.000	0.050
Observations	366	472	353	350

Notes: All specifications include utility and watershed $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

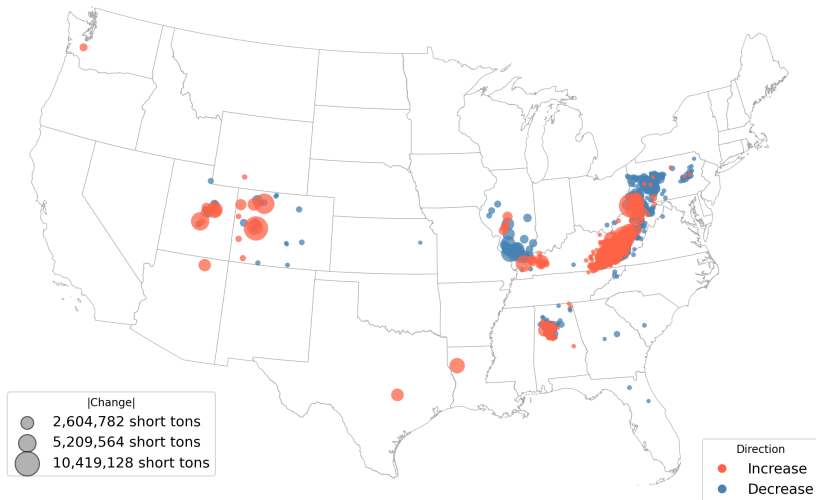
Per sample mean tons mined upstream: nitrate +0.076 mg/L (**), 9.4% of its mean; selenium +0.0003 mg/L (**), 6.9% of its mean. **Every reading in the sample stays under its MCL.** In the sample, only 49 IOC MCL violations.

Does mining increase under-reporting? Instrumenting coal mining

- OLS: mining and utility characteristics not random.
- Adapt Douglas and Wiggins (2015) Acid Rain Program (“ARP”) instrument for coal production.
- ARP set coal fired power plant SO_2 limits.
- Low sulfur coal was the main compliance response (Lattanzio, 2022).
- Use coal sulfur % and the 1995 SO_2 power plant compliance deadline.



Where coal production moved, 1985–2005



2SLS

First stage:

$$\text{NumMines}_{ct} = \alpha_0 + \alpha_1 X_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ + \text{State}_s \cdot Y_t + C_c + \nu_{ct}$$

Second stage:

$$\text{Outcome}_{vct} = \beta_0 + \beta_1 X_{ct} + \gamma \widehat{\text{NumMines}}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

- $\text{Post95}_t = \mathbf{1}\{t \geq 1995\}$. Sulfur is time-invariant, so C_c absorbs its level.
- X is time varying utility characteristics.
- Outcome: 1 if utility c had a violation, visit, or enforcement of type v at any point in year t .
- Coefficients are in percentage points.
- SEs clustered at the utility level.

Instrument validity

- **Relevance.** First stage F-stat 49.2.
- **Independence.** The ARP is a national clean-air law set by national energy and emissions politics, not downstream utilities' water quality.
- **Exclusion.** Effect of instrumenting mining downstream of the utility on utility behavior is not significant. (MR violations →), (visits →), (enforcement →).
Utility characteristics are not predicted by the instrument. (full balance table →)
- **Monotonicity.** A defier is a high-sulfur watershed that expanded *because* its coal was high sulfur. Scrubber adoption costs more likely lead to lost rather than gained market share; low-cost high-sulfur regions that expand anyway are never-takers.

Upstream mining reduces utilities' self-reporting

	Monitoring and reporting (MR) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal mines (sum)	0.68** (0.33)	0.55 (0.34)	0.62** (0.31)	0.42 (0.33)	0.49 (0.32)	0.19 (0.34)
2SLS						
Upstream coal mines (sum)	5.12*** (1.70)	3.94** (1.75)	4.85*** (1.57)	3.78** (1.55)	4.66*** (1.62)	3.06* (1.66)
RF						
Post-1995 $\times$ Upstream sulfur %	-2.12*** (0.66)	-1.95** (0.83)	-2.01*** (0.60)	-1.87** (0.73)	-1.93*** (0.63)	-1.51* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State $\times$ year fixed effects		✓		✓		✓

Notes: The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Base rates of nitrates, arsenic, and IOC are 5.7%, 3.7%, and 4.3%.

Scaling by 1st stage, a 1-SD decrease in sulfur leads to 0.56 additional mines after 1995, raising nitrate, arsenic, and IOC MR violations by 2.2, 2.1, and 1.7 pp.

The regulator increases inspections

	Sanitary	Technical assistance	Enforcement	Sample collection	Inspection
	(1)	(2)	(3)	(4)	(5)
2SLS					
Upstream coal mines (sum)	0.78 (1.20)	0.67 (0.95)	1.07 (0.81)	0.48 (0.50)	2.13** (1.07)

Notes: All specifications include utility and state $\times$ year fixed effects. The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Inspection visits rise significantly (+2.1 pp; baseline inspection rate 3.4%).

Scaling by 1st stage, a 1-SD decrease in sulfur leads to 0.56 additional mines after 1995, raising inspection visits by 1.2 pp.

Despite inspection increase, enforcement does not increase

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
2SLS						
Upstream coal mines (sum)	-3.06 (2.15)	-0.71 (2.10)	-3.37*** (0.95)	-1.08 (0.86)	2.37 (2.11)	-0.42 (2.25)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓

Notes: The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Baseline enforcement rates: informal 18.3%, formal 2.9%.

One more upstream mine reduces formal enforcement by 3.4 pp with utility and year fixed effects alone.

Scaling by 1st stage, a 1-SD decrease in sulfur leads to 0.47 additional mines after 1995, reducing formal enforcement by 1.6 pp.

Inspections would reveal concealment and increase penalties - which we do not see.

Conclusion

- Contamination reduces self-reporting.
- Contaminant data and increases in inspections without increasing enforcement suggests that under-reporting is not concealing MCLs.
- Evidence of cost driven under-reporting.
- Regulators' inspection visits substitute for utility self-reporting.
- MR violations are not strongly punished.
- **Policy:** raising self-reporting penalties or reducing self-reporting costs could reduce under-reporting.

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Appendix

Appendix: instrument balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.9942*** (0.3834)
Post-1995 $\times$ Upstream sulfur	0.0509 (0.0386)	0.0563 (0.0376)
Utility fixed effects	✓	
Year fixed effects		✓
State $\times$ year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

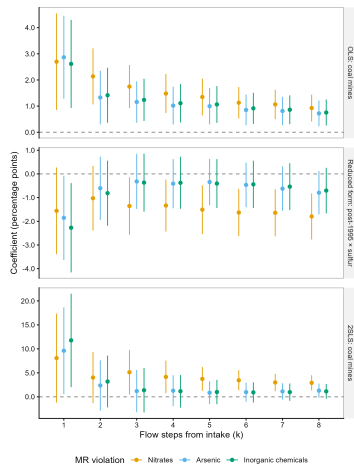
Appendix: balance test on utility characteristics

	Cumul. upstream coal prod. (10M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.2504 (0.2532)		-0.1596 (0.1198)	
Log backcast pop. 1997	0.1091 (0.0750)	0.1512 (0.0938)	0.0159 (0.0126)	0.0208 (0.0144)
Log pop. served (SYR2)	-0.0934 (0.0666)	-0.1113 (0.0956)	0.0060 (0.0259)	-0.0301 (0.0334)
Number of intake facilities	0.0089 (0.0569)	-0.0324 (0.0744)	0.0680* (0.0359)	0.0721* (0.0406)
Surface water source	0.1225 (0.1643)	-0.3902 (0.3296)	0.1210 (0.1307)	0.0521 (0.1601)
Public ownership	-0.1554 (0.2551)	-0.0599 (0.2240)	0.2084** (0.0911)	0.1048 (0.1098)
Wholesaler	-0.3780* (0.2041)	-0.7465 (0.9141)	-0.2164 (0.1340)	-0.2212 (0.1683)
Source water protection	-0.4549 (0.2749)	-0.4385 (0.4305)	-0.2807* (0.1521)	-0.1460 (0.1930)
Joint <i>F</i> -test (all covariates)	2.33	0.96	8.96	2.46
<i>p</i> -value	0.029	0.461	0.000	0.023
Observations	122	120	122	120
State fixed effects		✓		✓

Notes: Heteroskedasticity-robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

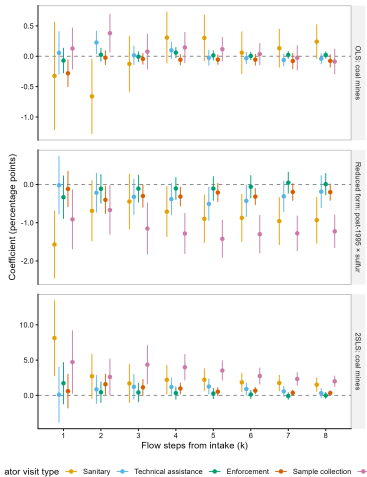
Observable characteristics jointly predict continuous upstream production only without state fixed effects ($p=0.029$) and lose significance once state fixed effects are added ($p=0.461$), as in the main specification. The joint test on the binary presence of upstream mining stays significant with or without state fixed effects.

Appendix: exclusion placebo, MR violations across k



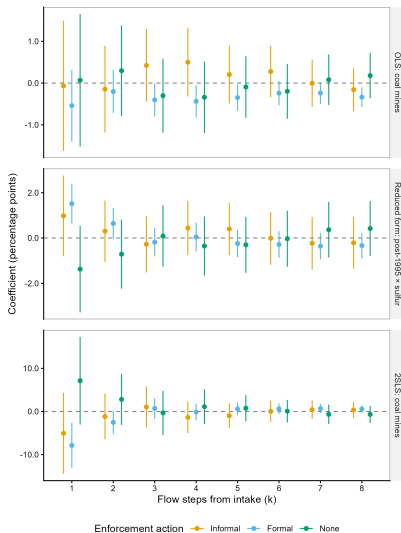
For utilities upstream of a mine — where mining cannot reach the intake hydrologically — OLS estimates on MR violations are positive and significant at every k . In the reduced form and 2SLS, nitrate estimates are significant for $k \geq 3$, while arsenic and inorganic chemical estimates are significant only at $k = 1$.

Appendix: exclusion placebo, visits across k



For utilities upstream of a mine, inspection visits rise significantly at nearly every k , sanitary visits at $k = 1$ and $k \geq 4$, and sample collection visits at several radii; technical assistance and enforcement visits are mostly insignificant.

Appendix: exclusion placebo, enforcement across k



Enforcement actions show no consistent pattern for utilities upstream of a mine at any choice of k , supporting the exclusion restriction.

References I



Douglas, Stratford and Seth Wiggins (2015). *Effects of Acid Rain Regulations on Production of Eastern Coals of Varying Sulfur Content*. College of Business and Economics Working Paper. West Virginia University.



Lattanzio, Richard K. (2022). *Clean Air Act: A Summary of the Act and Its Major Requirements*. Tech. rep. Washington, DC: Congressional Research Service.



Oleckno, William A. (1982). "The National Interim Primary Drinking Water Standards: Part II, Impact on Community Water Systems: The Indiana Experience". In: *Journal of Environmental Health* 44.5, pp. 240–244.